Chen Qian

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Summary

- Ph.D. candidate in Applied Mathematics at UC Davis (expected June 2027), with an M.S. in Statistics, specializing in **time series, statistical learning, and inference for dependent data**.
- Over 4 years of research experience in **time series**, developing methods for **time series classification and time-varying dependence estimation**, with **asymptotic analysis and empirical evaluation**; **first author** of a paper published in *JRSS Series B*.
- Research interests center on **modeling and inference for dependent data**, with an emphasis on **empirical performance and interpretability**. Interested in developing and applying mathematical and statistical methods to address **problems with real-world impact**.

Education

University of California, Davis **Sep. 2023 - June 2027 (expected)**
Ph.D. in Applied Mathematics *Davis, CA*
Research Advisor: Prof. Xiucai Ding, Department of Statistics, UC Davis

University of California, Davis **Sep. 2021 - June 2023**
M.S. in Statistics *Davis, CA*

Complutense University of Madrid **Sep. 2016 - July 2020**
B.S. in Applied Statistics *Madrid, Spain*

Publications

- **Qian, C.**, Ding, X., & Li, L. (2026). Structural classification of locally stationary time series based on second-order characteristics. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, qkag083. DOI

Manuscripts & Work in Progress

- Ding, X., **Qian, C.**, & Wang, J. Structural Approximations for Multivariate Locally Stationary Time Series with Applications to Multichannel Signal Analysis. *Completed manuscript; final proofreading.*
- Ding, X., **Qian, C.**, & Zhou, Z. On the Simultaneous Inferential Theory for Locally Stationary Time Series under Constraints. *Work in progress.*

Honors & Awards

- 2026 Conference Travel Award, Conference on Statistical Foundations of Artificial Intelligence, UC Santa Barbara (poster presentation forthcoming).
- 2026 L&S Graduate Summer Support Award, University of California, Davis (selected; declined due to COSMOS Teaching Fellow appointment).

Research Experience

Structural Classification of Locally Stationary Time Series Nov. 2024 - Apr. 2026

Joint work with Xiucai Ding and Lexin Li; JRSS Series B, 2026

- Developed a classifier for locally stationary time series using time-varying autoregressive models to extract interpretable features of temporal dependence.
- Established feature consistency and asymptotic classification guarantees under the class-separation conditions.
- Conducted simulation studies and implemented a classification workflow for normal/abnormal EEG recordings in the TUH Abnormal EEG Corpus; obtained 94.7% accuracy on 150 test subjects.

Structural Approximation of Multivariate Time Series and Multichannel EEG Classification Jan. 2026 - Present

Joint work with Xiucai Ding and Jeffrey Wang; completed manuscript

- Implemented estimation, cross-validated model selection, and classification algorithms; conducted simulation experiments across temporal dependence patterns and class-balance settings.
- Analyzed multichannel EEG data from the Siena Scalp EEG Database for seizure/non-seizure classification; correctly classified 61 of 64 test events (95.3% accuracy; F1 score 0.914).

Constrained Autocorrelation Estimation and Simultaneous Inference June 2026 - Present

Joint work with Xiucai Ding and Zhou Zhou; work in progress

- Formulated time-varying autocorrelation estimation for locally stationary time series with constrained sieve regressions; constructed its simultaneous confidence bands.
- Implemented a block multiplier procedure and evaluated confidence-band coverage through repeated-sampling simulations; working on proofs of uniform consistency for the autocorrelation estimator and asymptotic validity of simultaneous confidence bands.

Teaching Experience

University of California, Davis **2023 - Present**

Teaching Assistant *Davis, CA*

- Led discussion and office hours for courses in experiment design, data science, and calculus; prepared instructional materials and supported assessment.

University of California, Davis **July 2026**

COSMOS summer program Teaching Fellow *Davis, CA*

- Taught academically advanced high-school students core ideas in statistical modeling and exploratory data analysis.
- Mentored student teams in formulating research questions, selecting methods, interpreting results, and presenting technically sound conclusions.

Additional Professional Experience

Zhongyu Tech **Sep. 2020 - May 2021**

Data Scientist Intern, Mobile Game Analytics *Wuhan, China*

- Designed product experiments and conducted cohort, retention, and churn analyses for the mobile game *Wheel Offroad 3D*.
- Built ETL workflows and recurring analytical reports that translated gameplay data into recommendations for product and engineering teams.

Technical Skills

Programming: Python, R

Languages: Mandarin Chinese (native), English (professional), Spanish (working proficiency)